The Vent-Snap Ratchet: Quantifying, Modeling, and Eliminating Seal-Rupture Slip in Suction-Based Wall Climbing Without Force Sensing

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***Abstract*—Legged robots that climb with vacuum suction cups must break a seal at every step. On compliant, position-controlled platforms we show that this seal rupture releases elastic energy stored in the leg chains: the body drops within tens of milliseconds and the next adhesion locks the loss in place, so that a robot stepping in place with zero net command descends by a ratchet. Using a low-cost phone-video pipeline (normalized cross-correlation tracking, per-run metric scale, command-marker time synchronization), we quantify the effect at millimeter and per-leg resolution: 93% of the per-cycle position loss occurs within $100\,$ms of rupture, while during venting itself the body is stationary—so slower venting provably cannot help. A one-dimensional parallel-spring model explains the loss and makes four predictions that we verify, including strict linearity of the rupture bounce in a pre-laid unloading displacement. Building on this, we propose the *zero-force handover*: before venting, the lifting leg's foot command walks back a per-leg calibrated displacement δ_j while the supporting legs absorb it under a mean-invariant apportionment with rotation-distance weights—purely kinematic, open loop, and using no force, tactile, pressure, or current feedback. In an $n{=}3$ Latin-square study the method cuts total slip by 43% (δ alone) and 61% (with weights), and rupture-segment slip by $82\%/88\%$ (paired $p \leq 0.0022$). A pre-registered rate-discrimination experiment shows the residual transfer loss scales with laid millimeters, not elapsed time ($\Delta b = \kappa\,\delta_j$, κ reproduced to 4% across five days), identifying load-share allocation—not speed—as the remaining lever (κ reduced 44% by the weights). Data, logs, and the measurement pipeline are released.**

***Index Terms*—Climbing robots, legged robots, compliance and impedance control, calibration and identification, mechanism design.**

I. Introduction

LEGGED climbing robots that adhere by vacuum must do something wheeled and sliding-suction climbers avoid: at every single step they destroy an adhesion contact and rebuild it elsewhere. For negative-pressure feet the destruction is not gradual. A vented cup holds its grip while residual vacuum persists and then lets go in one instant when the seal lip peels—a *rupture-type* detachment that concentrates the mechanical consequences of releasing a loaded contact into a few milliseconds, far below the control bandwidth of any gait controller.

On a stiff robot this would matter little. But small wall-climbing platforms are not stiff: safety margins are bought with lightweight, 3D-printed leg chains, bellows cups that conform to the surface, and low-cost position-controlled servos with no torque feedback. On such a platform we observed that a hexapod commanded to *step in place* on vertical glass—lift each foot and put it back exactly where it was—slides down the wall by 57–74 mm per gait cycle, with zero interfacial slip of the attached cups and zero drift at rest. Frame-by-frame decomposition shows that in the $n{=}3$ study of this letter 93% of the loss occurs within $100\,$ms of seal rupture (83% in the initial discovery run, whose segmentation attributed more to post-bounce re-equilibration), while during active venting, before the seal lets go, the body moves less than $0.15\,$mm. The mechanism is elastic: as the body sags, each attached leg chain is wound up like a spring; venting a loaded leg cuts that spring at full tension, the body falls until the remaining legs' additional deflection catches it, and the next zero-preload adhesion locks the new position in. We call the resulting staircase descent the *vent-snap ratchet*. Servo current corroborates the picture: bus current climbs step-wise from $0.7\,$A to $\sim$$1.9\,$A over two cycles and does not relax—the legs are increasingly fighting each other.

The failure combination—compliant support chains, rupture-type detachment, and position-domain actuation that accumulates internal forces [1]—is generic to low-cost suction climbers, yet we found no prior quantification of the resulting whole-body position loss (Sec. II). Prior treatments of detachment disturbance measure foot forces or body vibration and manage them with force sensors in closed loop; the classical walking-machine literature unloads a leg before lift-off entirely in the force domain. Our platform has no force, tactile, or per-joint current sensing at all, and retrofitting it would cost weight, wiring, and complexity precisely where climbing robots can least afford them.

Instead we exploit the one fact that makes the problem tractable: an attached foot cannot slip, so *changing its position command changes force, not position*. The unloading condition "lift only a force-free leg" can therefore be translated from the force domain into the displacement domain and executed open loop: before venting leg j, walk its foot command back up-slope by a per-leg constant δ_j—the stored deflection, calibrated offline by linear extrapolation of the rupture bounce—while the supporting feet absorb the same displacement under weights that keep the six-leg command mean invariant, so the body is commanded to stay put. We call this the *zero-force handover*.

The contributions of this letter are:

1) **Discovery and quantification of the vent-snap ratchet.**

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To the best of our knowledge, the first time-resolved, per-leg, millimeter quantification of whole-body position loss at the instant of vacuum-seal rupture in a legged suction climber, with a reproducible low-cost video pipeline and released data. Prior reports of detachment disturbance are qualitative for body motion, or quantify foot forces, accelerations, or cavity pressure instead [2], [3], [4].

2) **A parallel-spring account with verified predictions.** A one-line elastic model explains the loss, predicts strict linearity of the rupture bounce in the pre-laid displacement (verified; per-leg slopes spread $4\times$ and furnish a stiffness estimate that agrees with an independent route, $r = 0.938$), predicts the dilution effect of load-share weights (-32% predicted vs. -31–33% measured), and—via a pre-registered discrimination experiment—yields a transfer-loss law $\Delta b = \kappa\, \delta_j (1 + 0.20(r-1))$ in which time plays no role.
3) **A sensor-free, open-loop, software-only remedy with n=3 evidence.** The zero-force handover converts pre-detachment unloading—known in force-sensing closed-loop forms [5], [6]—into the displacement domain: offline per-leg calibration, mean-invariant apportionment, rotation weights, no sensors of any kind in the loop. In a Latin-square n=3 study on glass it reduces total per-cycle slip by 43%/61% and rupture-segment slip by 82%/88% (all pairwise comparisons $p \leq 0.0022$, surviving Bonferroni correction).

All claims of priority are made "to the best of our knowledge" and are bounded carefully against the nearest prior work in Sec. II. Logs, videos, verification crops, and the analysis pipeline will be released **[TODO: package the open-data bundle; add repository URL]**.

II. Related Work

1) Platforms: Adhesive legged climbers divide by attachment principle into vacuum/negative-pressure feet [7], [8], [9], [10], dry/gecko-inspired adhesion [11], [2], [12], and grasping or spine-based feet [13], [6]; see surveys [14], [15]. This letter concerns the oldest line, discrete-gait suction feet, where each step must break and re-make a seal—a cost that sliding-suction designs [16], [17] avoid by paying for continuous pumping instead. Our platform is a low-cost, 3-DOF-per-leg hexapod in this lineage.

2) Detachment disturbance: what has been measured: That releasing an adhesive foot disturbs the rest of the robot is known qualitatively: Stickybot's designers report that with isotropic pads the large detachment force "produc[ed] oscillations that frequently caused the other feet to slip" [2], and a recent Chinese survey names "transient and large detachment forces transmitted to the trunk causing body shudder" an open problem [3]. What has been *quantified* is foot-level: peel-angle impedance control reduces normal detachment force [18], rotational peeling cuts pull-off force by 65.5% [19], and trajectory/force shaping reduces maximum detachment force by 28% and IMU-measured vibration by 82% [20]. We note explicitly that the 82% in [20] is an acceleration amplitude (m/s^2) and numerically coincides with, but is dimensionally unrelated to, our 82% reduction of rupture-segment *displacement* (mm). In the vacuum lineage the only transient quantification we found is of cavity pressure [4]. Whole-body position loss per detachment event—the quantity that actually bounds climbing progress—appears in no prior work we could locate.

3) Unloading before detachment: The idea of unloading a foot before releasing it has force-domain, closed-loop precedents that we must and do delimit against. The Stickybot patent describes tangentially unloading a foot before lift-off to relax accumulated force and elastic deformation, using Hall-effect force sensors in a stiffness-control loop [5]. LORIS's gait state machine contains an explicit *Unload* step that constrains the departing gripper's contact force to zero in an online force optimization [6]. LEMUR 3 manages attach/detach forces with a per-limb force sensor [13]; admittance control mitigates reaction forces on limbed climbers in simulation [21]; reaction-aware planning shapes swing inertia [22]; quasi-static gait optimization redistributes contact forces at the planning level [23]. All of these execute in the force domain, in closed loop, or at the planning layer, and none quantifies body slip. Our claim is accordingly narrow: converting the unloading condition to the *displacement domain* (per-leg offline calibration, mean-invariant apportionment), executing it *open loop on a platform with no force sensing whatsoever*, and validating it by *rupture-segment slip*.

4) Force distribution in walking machines: Smoothly driving a leg's force to zero before lift-off is a classical walking-machine topic from Waldron's force/motion management onward [24], [25], [26], [27], [28], through to modern whole-body controllers and MPC [29], [30], [31], [32]. The entire line executes in the force domain: either force/torque sensors close the loop [33], [1], or force setpoints are handed to actuators that can track them. Chen et al.'s "does not require force sensors" [34] refers to the setpoint computation only. The DLR Crawler paper states the problem we start from—position-controlled multi-legged robots accumulate internal forces—and solves it with joint torque sensing [1]. Insects solve it too: load transfer to neighboring legs mechanically unloads a leg, and load sensors trigger its swing [35]—a biological closed loop that our open-loop table replaces. In this classical frame, our contribution is a new execution domain for an old requirement, on a platform class (hobby position servos, strong series compliance) the classical solutions do not cover.

5) Displacement-domain feed-forward for compliance: Calibrate-then-feed-forward compensation of structural compliance is itself a mature method family: TITAN XI pre-calibrates leg flexibility to restore foothold accuracy on slopes [36]; Ota et al. compensate gravity-induced deformation on a suction quadruped for motion smoothness [37]; machining robots correct commanded trajectories from offline stiffness models [38], [39]; legged humanoids feed forward sole/joint deformation [40]; a gecko-inspired climber applies stiffness-model-based gravity compensation for posture [41]; sway compensation shifts the body to transfer *gravity* load in static gaits [42]. Same mathematical shape, different target: none of these uses calibrated displacement to *zero the internal*

[TODO: Fig. 1: (a) robot on glass wall, (b) leg chain schematic with compliance sources and suction foot cutaway, (c) handover kinematics sketch (lifting leg walks back δ_j up-slope, supports absorb $w_i\delta_j$ down-slope).]

Fig. 1. Platform and the zero-force handover.

force of a leg about to detach, and none faces energy release concentrated into milliseconds.

6) Alternative routes: One can avoid rupture altogether (sliding suction [17], at the price of continuous power), estimate contact force from motor current [43] (infeasible on hobby servos with large gearbox friction; see Sec. VII), or exploit snap-through energy release for propulsion instead of suffering it [44]—the same physics used in the opposite direction. Directional dry adhesives detach at near-zero force by material design [2], [45]; vacuum cups cannot inherit that trick, which is precisely why the stored energy must be *returned before rupture*. Slow venting, the intuitive remedy, is refuted by our data (Sec. III).

III. Platform, Pipeline, and the Phenomenon

A. Platform

The testbed (Fig. 1) is a 3D-printed hexapod derived from an open-source walking platform, with self-designed suction feet: each 3-DOF leg (coxa/femur/tibia $43/80/124$ mm) ends in a 30 mm bellows suction cup fed through a normally-closed 3-way solenoid valve and a per-foot check valve. Two diaphragm pumps hold the attached cups between -60 and -75 kPa by bang-bang hysteresis; adhesion is declared below -30 kPa; a leg may lift only when the five other cups are attached and deeper than a -50 kPa gate. Eighteen 35 kg·cm position-controlled hobby servos (no torque, force, tactile, or per-joint current feedback of any kind) are driven by a Raspberry Pi 5 through an RP2040 servo board; total mass is $\sim$2.7 kg **[TODO: final weigh-in of assembled robot]**. All experiments run on vertical glass with a safety rope. The platform's compliance is not incidental: printed legs, bellows cups, and servo gear trains together make the body-to-anchor chain soft enough that commanded body motion is only 42% realized (Sec. VI-D)—typical of the lightweight construction this robot class requires.

B. Measurement pipeline

We measure body position along the wall from a fixed phone camera (4K@30 fps) by normalized cross-correlation (NCC) template tracking of the robot's electronics board, with sub-pixel parabolic interpolation and frame-wise compensation of camera motion against a static reference patch. Metric scale is taken *per run* from the long edge (93.0 mm) of a PCB rigidly mounted on the robot, localized by four-corner sub-pixel fits (0.288–0.296 mm/px across runs; top/bottom edge disagreement 0.2–1.9%). Video time is registered to the robot's on-board event log by a commanded time-sync marker: a ±10 mm body lean executed at the start of each run and fit in shape to the tracked trajectory (R^2 0.95–1.00 per run); the same marker doubles as a compliance probe (Sec. VI-D). Every lift event is then decomposed into *handover* (displacement laying, when enabled), *vent* (a fixed $[-1, +0.4]$ s window around the logged pressure-zero event, containing the seal rupture), *swing*, *press* (touchdown), and *settle* segments. The pipeline was verified against a manual tape-ruler reading (131 mm total vs. 131.3 mm tracked) and 188 archived localization crops; all scale boards and template locks were human-checked **[TODO: exact vent-window definition: confirm -1 s/$+0.4$ s bounds from ab_quant before submission]**.

C. The vent-snap ratchet

Stepping in place is the cleanest probe: each leg in the fixed rotation order (R3→L1→R2→L3→R1→L2) is lifted 33 mm, hovers 1 s, and is put back at exactly its former station; the commanded net body motion is zero, so any systematic displacement must come from elastic strain release and re-locking. Two controls verify the premises: ink marks on the glass beside each attached cup do not move (zero interfacial slip), and the body is stationary during all-feet attached dwell (zero time-type creep; see also Sec. VI-E). Yet the body descends 57–74 mm per cycle (discovery run; 63.9 ± 0.9 mm in the n=3 baseline, Sec. VI-B), in a staircase whose every step coincides with a seal rupture (Fig. 2).

Three time scales matter. (i) While residual vacuum drains after the venting command, the body does not move ($<$ 0.15 mm). (ii) At seal rupture the entire drop completes within 1–2 frames (33–67 ms), with an underdamped overshoot (L1: peak 23.8 mm, settle 21.5 mm)—a textbook elastic release. (iii) Between events the trajectory is flat. In the n=3 baseline the vent segments sum to $172.9{\pm}3.4$ of $185.5{\pm}3.0$ mm total—93%; touchdown press contributes 4% (discovery-run decomposition: $83\%/4\%$ with wider re-equilibration windows). The released foot itself recoils 9–15 mm down-wall at the same instant. Per-leg single-event bounces reach 20.9 mm (L1) and 19.2 mm (R1)—the top (front) legs, where peel moments concentrate—versus 5.8 mm for the softest legs, a first sign that leg-chain stiffness is strongly unequal.

Two corollaries fix the design space. First, *slow venting cannot work*: the force release is not distributed over the pressure decay but compressed into the lip's peel instant; the data show zero motion while residual vacuum drains. Any remedy must ensure the leg carries no force *at the moment the seal lets go*. Second, the loss is locked in by the next adhesion: a fresh cup grips wherever the body then hangs, at zero preload (Sec. IV), so the drops accumulate as a ratchet rather than averaging out. Servo bus current tells the same story electrically: from 0.73 A at first attachment it climbs step-wise to $\sim$1.9 A after two cycles and does not relax—each locked-in drop leaves the support legs fighting a larger internal force.

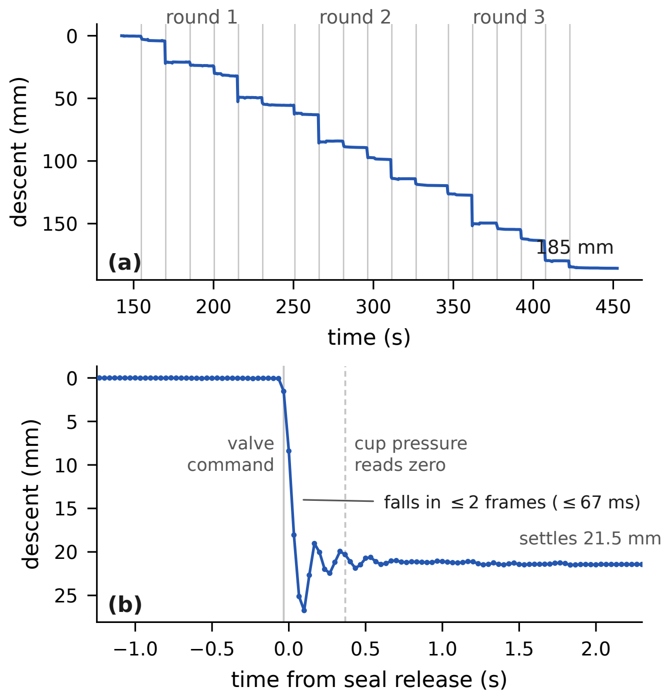

Fig. 2. The vent-snap ratchet (baseline, replication #1). (a) Body position along the wall over three rounds of stepping in place: every step of the 185 mm staircase coincides with a seal rupture (gray lines); the trace is flat between events. (b) Native-30 fps profile of one lift (L1, round 2), t=0 at the tracked release: the body is stationary before the release, falls 21.5 mm within two frames with an underdamped overshoot (peak 26.7 mm); the released foot simultaneously recoils 12.8 mm down-wall (not shown). The logged valve command and pressure-zero events bracket the mechanical release; their placement relative to the video carries the sync uncertainty (a few tenths of a second).

IV. A Parallel-Spring Account

A. Model

Collapse each attached leg chain i to a linear spring of stiffness k_i along the wall's gravity axis (x positive down). Let a_i be the (immovable) anchor of foot i and c_i its commanded position in the body frame, and define $y_i \equiv a_i - c_i$: the body position at which leg i would carry zero force. Leg i then exerts the shear force $f_i = k_i\,(x - y_i)$ on the body (positive up-wall), and static equilibrium of the attached set S gives

$$x \;=\; \langle y\rangle_k + \frac{mg}{K}, \qquad \langle y\rangle_k \equiv \frac{\sum_{i\in S} k_i y_i}{K}, \quad K \equiv \sum_{i\in S} k_i. \quad (1)$$

The body hangs below the *stiffness-weighted* mean of the legs' zero-force positions. Three events drive the dynamics:

Zero-preload lock-in. A cup that seals grips wherever the body currently is, storing no force: adhesion sets $y_j \leftarrow x$.

Rupture. Venting leg j removes a spring carrying tension f_j; the body falls to the new equilibrium of the catching set,

$$b_j \;=\; \frac{f_j}{K - k_j}, \quad (2)$$

i.e. *the observed bounce is the leg's stored force divided by the parallel stiffness of the other legs*—not the leg's own stored deflection $e_j = f_j/k_j$. The ratio $e_j/b_j = (K - k_j)/k_j$ is 5–8 on our platform, a distinction that matters for calibration (Sec. V-B).

Winding. Between adhesion and lift, every millimeter the body descends winds each attached leg by one millimeter ($x - y_i$ grows); older anchors are wound tighter, so the legs' stored forces differ even under identical commands.

Iterating lock-in, winding, and rupture over the gait reproduces the ratchet: in the 1-D simulation the steady-state in-place loss is $\approx 0.4\, mg/\bar{k}$ per cycle, and matching the observed 74 mm/cycle gives a whole-robot compliance scale of $mg/\bar{k} \approx$ 185 mm. The measured single-event bounce saturates near 21 mm, below the linear prediction—servo torque saturation caps the energy a single leg can store, which also explains why forward climbing still nets +26% of commanded stride rather than losing ground **[TODO: net-advance demo (T4b) will refresh this number]**.

B. Four verified predictions

P1: The rupture bounce is strictly linear in a pre-laid unloading displacement, with slope encoding leg stiffness. If, before venting, y_j is walked toward x by δ (Sec. V), the residual tension falls by $k_j\delta$, so $b_j(\delta) = b_j(0) - s_j\,\delta$ with $s_j = k_j/(K - k_j)$. A three-dose calibration experiment ($\delta = 0/12/24$ mm uniform; Sec. V-B, Fig. 3) finds all six legs strictly linear (mid-dose residual ≤ 1 mm), with slopes 0.13–0.57—a $4\times$ per-leg spread that directly measures the stiffness imbalance. The normalized stiffnesses $\hat{k}_j = s_j/(1 + s_j)$ from these slopes agree with a completely independent second route (fitting handover-segment sag, Sec. VI-E) at Pearson $r = 0.938$.

P2: A freshly attached leg bounces by nothing. Zero-preload lock-in implies the first lift of a just-attached leg releases only the force accrued since lock-in: measured first-round first-lift bounces are ≈ 0 (-0.0 and $+1.8$ mm in two independent runs), rising in later rounds as winding accumulates.

P3: Equilibrium is stiffness-weighted, so mean-invariant commands do not perfectly freeze the body. By (1), a handover that moves y_j up by δ and the supports down by weights w_i ($\sum w_i = 1$) shifts equilibrium by

$$\Delta x \;=\; \frac{\delta\,\left(k_j - \sum_{i\in S\setminus j} k_i w_i\right)}{K}, \quad (3)$$

zero only for equal stiffness. Unloading the *stiffest* leg must sink the body; the first-order prediction for L1 is 4.6 mm vs. 6.8 mm measured (first round), and the full fitted form of (3) reproduces all 18 leg$\times$round cells of the B condition with RMSE 0.85 mm (Sec. VI-E).

P4: Load-share weighting dilutes stored energy non-monotonically. Because each lock-in re-zeros one leg, load handed to a support early is partially re-absorbed by subsequent lock-ins, while load handed late is carried intact into that leg's own lift. Weighting the handover shares toward the legs *farthest from their next lift* (most dilution opportunities ahead) lowers steady-state stored energy: the 1-D model gives −19% (mild), −32% (the deployed schedule 0.6/0.25/0.1/0.05/0), 0% (all-to-one; non-monotonic), and +30% (reversed—worse than uniform). The measured B→C reduction of residual slip is −31–33% across metrics (Sec. VI-B).

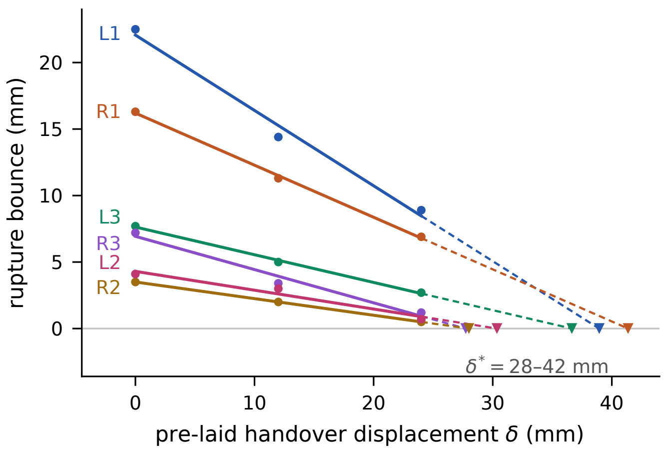

Fig. 3. P1 verified: rupture bounce vs. pre-laid displacement δ (round 2 of the three-dose calibration, $\delta = 0/12/24$ mm uniform). All six legs are strictly linear (mid-dose residual ≤ 1 mm); the slopes spread $4\times$ across legs (stiffness imbalance), and the dashed extrapolations to zero bounce define the calibration targets $\delta_j^* = 28$–42 mm (triangles).

V. Zero-Force Handover

A. Displacement-domain unloading

The remedy follows from one invariant: an attached cup does not slip (verified, Sec. III-C), so foot commands move forces, not feet. Before venting leg j, the gait engine inserts a HANDOVER phase: the lifting leg's foot command walks *up-slope* by δ_j (returning its stored deflection), while each supporting foot command walks *down-slope* by $w_i \, \delta_j$ with $\sum_i w_i = 1$ (absorbing the load), both laid quasi-statically at 10 mm/s along the gravity direction maintained by the engine. The six-command mean is invariant, so the body is commanded to stay put; the lifted leg's force is walked to zero; the seal then ruptures with nothing to release. The supports never carry more than they would carry *after* the lift anyway—the handover only moves the transfer earlier and makes it lossless. Commands remain bounded: over one gait cycle a support absorbs $\approx\delta$ down-slope in shares and returns $+\delta_j$ when its own turn comes, so offsets do not diverge (with per-leg tables the touchdown reset absorbs the small residual; the first-round command artifact this causes is bounded at $\sim$5 mm and vanishes in steady state).

B. Per-leg calibration: bounce is not stored energy

The naive calibration—set δ_j to the observed bounce—fails by the factor in (2): the bounce is the stored force read through the *other* legs' parallel stiffness, and underestimates the stored deflection 5–8$\times$. (Our first table, bounce $\times 0.8$, reduced per-cycle slip only $74 \rightarrow 50$ mm at full dose.) The correct, scale-free procedure drops out of P1: run three doses ($\delta = 0$, half, full), fit the per-leg line, and extrapolate to the zero-bounce intercept $\delta_j^* = b_j(0)/s_j$. Measured $\delta^* = 28$–42 mm per leg; we deploy $\delta_j = 0.8\,\delta_j^*$ (under-shoot discipline: a small residual bounce is benign, overshooting preloads the leg in reverse), giving the table L1/R1/L3/R3/R2/L2 $= 31/33/29/22/22/24$ mm. Because δ^* is a ratio of same-unit quantities, it is independent of the video scale calibration—the most robust number in this letter. Doses respond cleanly ($65.6 \rightarrow 50.0 \rightarrow 38.7$ mm per cycle across $0/12/24$), and the staircase character is unchanged: δ lowers each step's height, not its timing.

C. Rotation-distance load-share weights

P4 fixes the apportionment: shares are assigned by *rotation distance*—the support that just lifted (farthest from its next lift) receives the largest share, the next-to-lift receives none ($0.6/0.25/0.1/0.05/0$ in rotation order). The engine renormalizes over the current support set each tick and falls back to uniform shares (with a logged note) if the lift order ever deviates from the rotation—the weights assume the cyclic gait, and misapplying them would pile load onto one leg. Weights change the steady-state operating point, so the δ table is calibrated *under* the weight schedule (Sec. VI-A).

D. Implementation and guards

The method is $\sim$200 lines in the gait engine: one new phase between the lift decision and VENT, one per-tick displacement update, no new hardware, no sensing, no per-robot tuning beyond the δ table. Safety guards (none of which fired in any run reported here): the laying pauses during leak-rescue; every per-tick target is envelope-checked and the handover is truncated (partial unloading, logged) rather than driving a support out of workspace; and the -50 kPa lift gate is rechecked after laying, since seconds have passed since the lift decision. Handover adds $\delta_j/10$ mm/s ≈ 2.2–3.3 s per step at the deployed table—the price in cycle time; the energy price is quantified in Sec. VI-F.

VI. Experiments

A. Protocol

Conditions. A: baseline stepping in place; B: A + zero-force handover with the calibrated table ($\delta_j = 0.8\delta_j^*$); C: B + rotation-distance weights (table calibrated under weights). One variable changes per condition.

Design. n=3 full-protocol replications, each an independent mount–run–dismount of all three conditions, ordered by a 3×3 Latin square (#1 A→B→C, #2 B→C→A, #3 C→A→B) so that condition is decoupled from within-session order, battery state, and glass condition. Each run: one keypress launches a fixed automated sequence—60 s settle, ± 10 mm time-sync marker, then 3 rounds $\times$ 6 legs of lift–hover–replace with pressure-gated pacing (next lift waits for pump-off at -75 kPa), 15 s between rounds, 30 s final settle—so pacing is identical across runs by construction. Battery ≥ 7.9 V at each session start; camera untouched within a session; scale re-measured per run. All nine logs show zero freezes, zero envelope truncations, zero gate waits. Judgment uses round 2 by protocol (round 1 contains the known ramp-up and command-reset artifacts; round 3 is the replicate check).

TABLE I
MAIN RESULTS, n=3 LATIN SQUARE (MM; MEAN $\pm$ SAMPLE SD OVER REPLICATIONS). PERCENTAGES VS. A.

	A (baseline)	B ($+\delta$)	C ($+$weights)
Round-2 slip	63.9 ± 0.9	38.1 ± 1.0	26.5 ± 1.7
3-round total	185.5 ± 3.0	105.6 ± 2.2 (-43%)	72.9 ± 3.0 (-61%)
Rupture segment	172.9 ± 3.4	31.6 ± 1.0 (-82%)	20.1 ± 3.3 (-88%)
Handover segment	—	60.2 ± 2.2	39.7 ± 4.3
Current rise (A)	$+0.97 \pm 0.02$	$+2.40 \pm 0.08$	$+1.85 \pm 0.08$

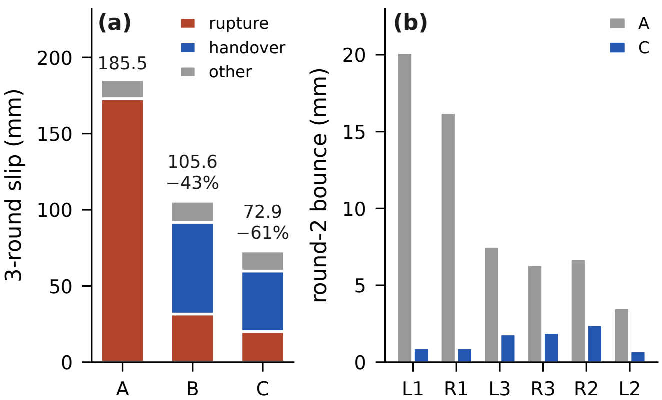

Fig. 4. Where the loss lives (n=3 means). (a) Three-round slip by segment and condition: the handover converts an impulsive 173 mm rupture loss into a 60 mm quasi-static transfer loss; weights compress it to 40 mm. (b) Per-leg round-2 rupture bounce, A vs. C: the stiff front legs drop from 20.1/16.2 to 0.9/0.9 mm.

B. Main results

Table I and Fig. 4 are the headline. Paired per-replication differences (df $=$ 2, two-sided): B$-$A $= -79.9 \pm 5.0$ mm ($t = 27.9$, $p = 0.0013$); C$-$A $= -112.6 \pm 5.9$ mm ($t = 32.9$, $p = 0.00092$); C$-$B $= -32.7 \pm 2.7$ mm ($t = 21.4$, $p = 0.0022$). All three survive Bonferroni correction for the three comparisons ($p \times 3 < 0.007$). We emphasize what n=3 means here: three *full-protocol* independent replications, each containing 18 lift events per condition, with relative effects nearly constant across replications (B/A: $-42/-45/-42\%$; C/A: $-61/-63/-58\%$)—the treatment effect is an order of magnitude larger than every noise scale in the experiment. No order effect is visible after the Latin-square decoupling.

Per-leg rupture bounces (round 2, n=3): the baseline reproduces the discovery run five days earlier (A-condition L1 20.1 ± 1.3 vs. 20.9); under C all legs are $\leq$2.4 mm, and the stiff front legs that dominated the baseline are 0.9 ± 0.5 (L1) and 0.9 ± 0.3 mm (R1)—a 95% per-leg reduction where it mattered most. R1 showed occasional slight upward bounce (-1 to -3 mm), the signature of mild overshoot, accepted per the under-shoot discipline.

C. Finding 1: the residual lives in the transfer, on the stiffest legs, and grows

With rupture slip suppressed, 56–62% of the B/C residual sits in the *handover segments*, concentrated on the two stiffest legs and growing round over round (B-condition L1: $6.8 \rightarrow 10.3 \rightarrow 12.6$, $6.9 \rightarrow 8.6 \rightarrow 10.6$, $6.7 \rightarrow 8.8 \rightarrow 11.9$ mm across the three replications; R1 similar; the other four legs 0–1.5 mm). The first-order reading—P3's stiffness-weighted equilibrium—is correct about the *shape* but, as the modeling below shows, wrong about the *size*: redistribution explains only $\sim$7 mm of the 60 mm segment. The decomposition of this residual, and which lever actually moves it, is the subject of Sec. VI-E.

D. Finding 2: a 42% command-to-body transfer ratio

The $\pm$10 mm time-sync marker doubles as a compliance probe: across 9 markers, 3 days, and 3 camera setups, the commanded 10 mm body lean produces 4.22 ± 0.30 mm of actual body motion—a *transfer ratio of* $42.2 \pm 3.0\%$, invariant across conditions (A/B/C 4.15/4.28/4.22 mm), as expected for a probe executed at full static support. Sixty percent of commanded displacement is absorbed by the leg chains. We report this as a platform characterization, not new physics: it quantifies the compliance that makes the ratchet large, and it feeds the model's overall gain (the fitted c_0 below).

E. Mechanism: fits, and a pre-registered discrimination experiment

1) Fitting the handover segment: Per-leg, per-round handover-segment displacements (B: 54 cells over 3 replications) are fit by

$$\Delta b_{j,r} \;=\; c_0\, \delta_j \big(\hat{k}_j - \textstyle\sum_i \hat{k}_i s_i\big)\big(1 + \gamma(r{-}1)\big) \;+\; \Delta b^{\mathrm{cm}}_{j,r}, \quad (4)$$

the first term being (3) with a round-growth factor γ and the second a *common-mode* term shared by all legs. Within B, the model achieves RMSE 0.85 mm on signals of 6.8–12.6 mm with leave-one-replication-out CV of 0.99 mm, and the fitted $\hat{k}$ agrees with the slope route of P1 at $r = 0.938$ (final joint estimate: L1 0.265, R1 0.227, L3 0.137, R2 0.133, R3 0.125, L2 0.112; $\gamma = 0.11$). The account it returns is a reversal of the first-order reading: B's 60.2 mm segment $=$ $\sim$6.7 redistribution $+$ $\sim$55.5 common-mode; C's $39.7 = 7.7 + 30.0$. *Ninety percent of the residual is a common-mode descent incurred per handover*—and the apparent concentration on stiff legs is a shape effect: on soft legs the redistribution term of (3) is *negative* (unloading a soft leg lifts the body slightly) and cancels the common-mode base, while on stiff legs the two add.

2) What is the common mode? A pre-registered discrimination: Two hypotheses fit the n=3 data equally well, because laying time and laid displacement are perfectly collinear at fixed rate ($T \approx 0.5 + \delta_j/10$): (H1) *time creep* ($\Delta b^{\mathrm{cm}} = \rho T$, loss $\propto$ window duration) vs. (H2) *per-millimeter transfer loss* ($\Delta b^{\mathrm{cm}} = \kappa\, \delta_j$). Model selection on observational data alone is provably helpless here—indeed on the n=3 data H1 narrowly "won" the information criterion, a collinearity masquerade. The only discriminator is intervention: change the laying *rate*. We pre-registered the experiment in the lab protocol with interpretation tables for both outcomes before touching the wall: three same-day C-vs-D′ pairs in alternating order, D′ identical to C except laying at 20 mm/s (measured window $3.18 \rightarrow 1.83$ s), predictions H1: handover segment

TABLE II
RATE DISCRIMINATION (D′): PREDICTIONS VS. OUTCOME (HANDOVER-SEGMENT SLIP PER RUN, MM), AND JOINT MODEL SELECTION OVER FOUR CONDITIONS (B/C/C_{31}/D′, 216 CELLS).

	prediction	measured
H1 time creep (ρT)	26.0	36.6 ± 1.6
H2 per-mm ($\kappa\delta$)	≈ 39.5	

Joint fit (ΔAICc; lower is better): **M10** $\kappa\delta$: **0.0**; M8 ρT: 17.8; per-condition constants: 31.6; current-based: 70.6/73.0.

39.7 → ∼26 mm (and ∼12 mm/run of free performance); H2: unchanged.

The outcome (Table II, Fig. 5) is unambiguous: doubling the rate leaves the handover segment at 36.6 ± 1.6 mm (paired D′−C $= -2.9 \pm 2.1$ mm, $p = 0.14$; totals -1.7 ± 0.7 mm, -3%), far from H1's 26.0; the landing point sits at 0.21 on the H1–H2 axis, and the small paired difference is itself largely window attribution (a ∼1 mm relaxation tail crossing into the vent window at the faster rate; the merged-segment comparison is stable). Aligning every handover on its vent instant (Fig. 5) shows the mechanism directly: the D′ ramp is twice as steep and the *plateau is identical* (L1 7.9–8.8 mm across all six runs)—the descent tracks laid millimeters quasi-instantaneously and does not accumulate with window time. Sanity checks confirm no new physics entered at 20 mm/s: per-leg rupture bounces, transfer ratio (4.30/4.42 mm), and current increments are unchanged between conditions.

3) The transfer-loss law and its lever: The joint four-condition fit (M10) settles the common mode as

$$\Delta b^{\mathrm{cm}}_{j,r} \;=\; \kappa\,\delta_j\,\bigl(1 + 0.20\,(r{-}1)\bigr), \tag{5}$$

with κ a property of the *load-share allocation*, not of time: uniform shares $\kappa_B = 0.099$ mm/mm; rotation weights $\kappa_C = 0.055$ (same-condition re-measurement five days later: 0.053—a 4% reproduction). The weights, introduced to dilute stored energy (P4), turn out to act mainly by *compressing* κ *by* 44%. Mechanistically the law points to a quasi-static per-millimeter loss at the cup/leg-chain interface during load transfer (micro-slip or structural nonlinearity)—categorically not time creep, consistent with the zero-drift dwell control. The practical reading: laying speed buys nothing (the rate parameter is kept only as a validated-harmless option); share allocation is the remaining lever; and the residual handover segment ∼37–40 mm is the intrinsic price of pre-laying $\sum_j \delta_j = 161$ mm per round at the current κ ($\kappa \sum \delta \times 3 \approx$ 26 mm + redistribution ∼8 + round growth).

F. Cost

The handover converts an impulsive loss into sustained servo load: current increase over a run rises from $+0.97 \pm 0.02$ A (A) to $+2.40 \pm 0.08$ A (B), which the weights claw back to $+1.85 \pm 0.08$ A (C, -23% vs. B—direction and roughly two-thirds of the modeled -32% stored-energy dilution). Cycle time grows by the laying duration (2.2–3.3 s per step at 10 mm/s). Both costs are the explicit price of zero sensing; Sec. VII argues the trade.

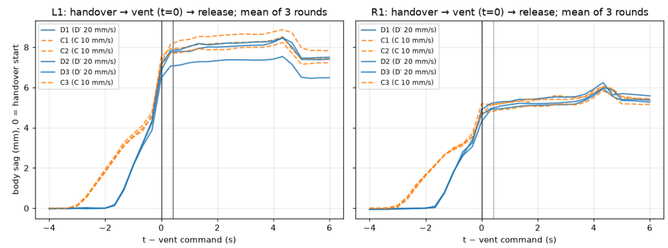

Fig. 5. The discrimination result, seen directly: handover profiles of all six runs aligned on the vent command (t=0), per-leg means over rounds (left L1, right R1). At double laying rate (D′, solid) the descent ramp is twice as steep as the control (C, dashed) but ends at the *same* plateau—the descent tracks laid millimeters, not elapsed time. **[TODO: restyle to match the other figures (fonts/colors); currently the analysis-report rendering]**.

G. Deviations and threats to validity

We record all deviations. (i) One run's video started late, cropping the time-sync marker; the initial rescue (periodic-comb step regression) locked one tooth off and mis-attributed per-leg slip; the error was caught by physical implausibility of the per-leg pattern, re-synchronized by a crop-tolerant marker fit ($R^2 = 0.99$, amplitude independently matching the other eight markers), and the corrected run agrees with all cross-checks. Lesson embedded in the pipeline: check comb periodicity before step-based sync; the 60 s pre-run settle is now built into the launch sequence. (ii) The camera moved between two sessions (never within one); metric scale is per-run by design, and in-run drift is compensated by the static reference (≤4.4 px). (iii) Battery was not recharged between runs within a session (7.5–8.1 V range); the Latin square decouples it from condition. (iv) NCC template score medians (0.39–0.86) fall below our internal 0.8 bar in tilted-board runs; the reference-patch score (≥ 0.988) and nine-run cross-consistency (CV 1.6–4%) bound the impact. (v) Scale is referenced to the robot-mounted board plane; top/bottom-edge disagreement (0.2–1.9%) bounds the residual perspective error. (vi) Segment boundaries at ±0.3 s move ∼1 mm between adjacent columns in two runs (relaxation tails); merged segments and totals are unaffected. None of these approaches the order of the effects (43–88%).

VII. DISCUSSION

Is this just repairing a badly built robot? The compliance is not a construction accident to be engineered away: lightweight printed legs, bellows cups, and unsensed gear trains are how sub-3 kg climbers are built at all, and every element of the failure combination—compliant support chains, rupture-type detachment, position-domain actuation—recurs across the platform class (Sec. II). What the analysis adds beyond the fix is a design rule: compensability is bounded by the support set's stiffness profile. The redistribution term (3) vanishes under no weighting for the stiffest leg (L1's residual coefficient is 0.019 but cannot reach zero)—so a designer who wants software-only detachment management should bound the *spread* of leg stiffnesses, not just their level. Conversely, because the remedy is a ∼200-line, zero-hardware retrofit, it is deployable on the installed base of position-controlled climbers as-is.

Why open loop, when the bus current plainly sees the load? We argued this choice from data rather than taste. Whole-robot current cannot be attributed per leg: during a stiff-leg handover the total shows a plateau whose onset is the balance point of unload/load gains, not the zero-force point (the leg still bounced 6.9 mm at a dose that "looked" converged); during a soft-leg handover the total rises monotonically—there is no signal to servo on. Per-leg force sensing is exactly the hardware bill we set out to avoid; motor-current force estimation [43] needs friction models hobby gear trains do not honor. Meanwhile the open-loop table is estimation-free, tuning-free, and—as the n=3 spread shows (relative effects within 3 points across replications)—stable across days without recalibration. The natural closed-loop upgrade is event-based, not continuous: the current step at the rupture instant (and ultimately an IMU pulse) is proportional to residual stored energy and would let the robot re-trim its own δ table between climbs; we deliberately leave this as future work so that the present result stands on zero sensing.

Model-gated experiment selection. The fitted model vetoed one of our own planned conditions: stiffness-aware shares (choosing w_i to zero (3) per leg) move only the redistribution term, worth 3–4 mm—not a wall experiment. It also priced the rate intervention correctly and pre-committed both interpretations. We found this discipline—models must state their number before the robot climbs—as valuable as any single result.

Generalization and limits. The phenomenon needs three ingredients: series compliance between anchor and body, detachment that releases the contact faster than the support can re-equilibrate, and actuation that tracks position, not force. Vacuum cups on any smooth surface qualify; directional dry adhesion largely does not (peel-by-design releases at near-zero force [45], [2]), which is why the gecko lineage never needed this letter. Our evidence is one platform, one surface, an in-place proxy task, three rounds, and n=3: the effects are huge and internally consistent, but absolute stiffness (a hang-weight calibration), net-advance climbing with the handover engaged, longer runs (round-growth factors γ, β' unsaturated at r=3), and the micro-mechanism of κ (share-concentration vs. cup freshness, separable by a share-perturbation experiment) remain open **[TODO: fold in T4a/T4b results when available]**. The δ table is an open-loop calibration: payload or posture changes outside the calibrated regime require re-running the three-dose procedure (one session) or the event-based re-trim above.

VIII. Conclusion

Every discrete-gait suction climber pays a tax at the moment its seals break. On a compliant, position-controlled hexapod we measured the tax precisely—93% of a 64 mm/cycle in-place descent concentrated into post-rupture milliseconds—traced it to elastic energy stored by the robot's own weight in its leg chains, and showed that the intuitive remedy (vent slower) cannot work even in principle. Because attached feet cannot slip, the cure fits in the command stream: return each leg's stored deflection before venting it, hand the load to the remaining legs under mean-invariant, rotation-weighted shares, and the seal breaks with nothing to release. With zero sensors, zero new hardware, and one offline calibration, the ratchet's rupture component falls by 88% and the total by 61%; what remains obeys a clean per-millimeter transfer law whose coefficient is set by load-share allocation—the lever we pull next.

Multimedia

[TODO: Attachment video (mandatory for this venue): A-vs-C side-by-side in-place runs + 30 fps slow-motion of a single rupture; source MOVs frozen on the lab machine; ≤20 MB/180 s if repurposed for conference submission.]

Data Availability

[TODO: Release bundle: nine + six run logs, trajectories and decomposition reports, 188 verification crops, analysis pipeline (tracking, sync, decomposition, fitting), and the δ-calibration protocol. Decide host (repo/Zenodo) and license.]

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